DE.AUTO

CHAPTER 10

TIRES & WHEELS

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

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DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Ten: Tires and Wheels

“FOUR PATCHES. THAT'S ALL YOU'VE GOT.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

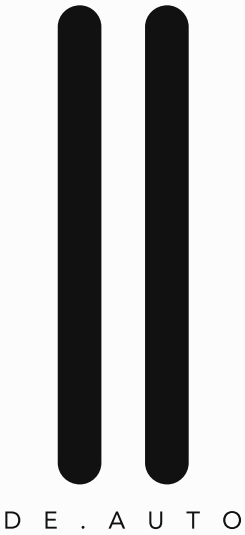
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

A lot of car people say: the best “modification” you can make to your car... is a tire that grips the ground properly. 🛞 Why do they say that?

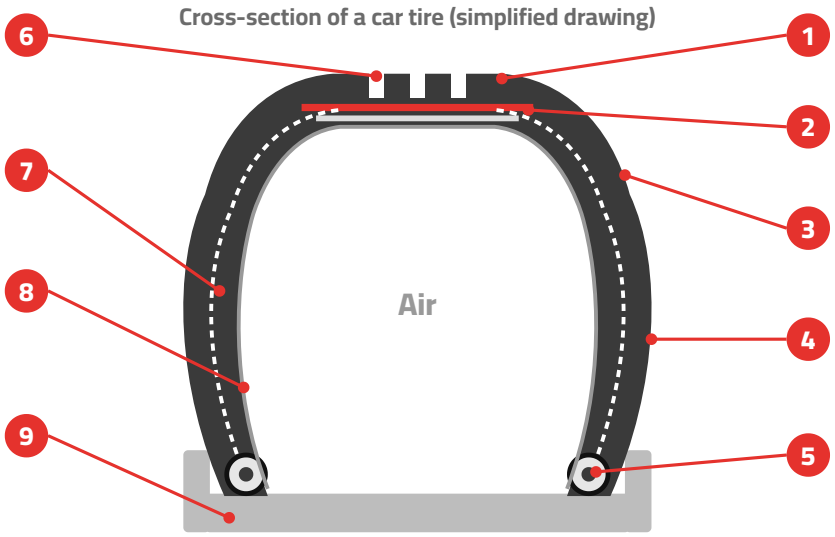
By the end of this chapter you'll know:

- ✓ What a tire is made of inside, and what each layer does
- ✓ Tire types (summer, off-road, run-flat, track...) and when each one suits you
- ✓ How to decode things like 98Y and UTQG, and how to read the manufacture date
- ✓ Why Gulf heat changes your tire pressure, and what wear patterns tell you
- ✓ The Camber / Toe / Caster angles in drawings, and balancing vs. alignment
- ✓ Offset and PCD and centre bore, without the headache
- ✓ Upsizing maths, speedometer error, and the risks of spacers and stretched tires



What's inside a tire? 🧅

A tire isn't just a "rubber ring". It's an **engineered sandwich** of layer upon layer: rubber that grips the ground, wires and fibres that hold the shape, and air that carries the weight. Here's a cross-section:



The red numbers match the cards below

1. Tread
The rubber that touches the tarmac, with the grooves in it.

2. Belts
Layers of steel wire under the tread that hold its shape and strengthen it.

3. Shoulder
The edge between the tread and the sidewall; it works hard in corners.

4. Sidewall
Carries the load and absorbs bumps. Your size and codes are written on it.

5. Bead (steel ring)
A ring of wire that locks the tire to the wheel and seals in the air.

6. Grooves
Channels that drain water and give the rubber room to move.

7. Carcass (plies)
Cords (fabric or steel) that carry the load; the basis of its strength.

8. Inner liner
Solid rubber that holds the air, in place of the old inner tube.

9. Rim flange
The place the tire bead sits on.

> 💡 **Did you know?** Modern tires are **tubeless**: the air is sealed between the inner liner and the rim, which is why a damaged wheel can leak even when the tire is fine.

Tire types: choose for the job 🎯

There's no tire that's "best at everything". Each type is designed for a job, and every advantage has a price. These are the general types:

Type	Advantage	The price you pay
Summer / performance	Sticky rubber and a stable sidewall: strong grip on dry roads and a precise feel through the steering wheel.	Suffers in the cold, usually wears faster, and is less comfortable.
All-season	A balance of comfort, life and everyday use.	Doesn't excel in any extreme condition; it gives you "the middle".
All-terrain	A chunky tread and tougher sidewall for dirt, gravel and light sand.	Louder, less grip on wet tarmac, and higher fuel consumption.
Run-flat	A reinforced sidewall that lets you drive a short distance at limited speed after losing air.	Harsher and more expensive, and you need a pressure-warning system that works properly.
Track / semi-slick	Very sticky rubber for tracks and organised events.	Wears out fast, is weak in the rain, and some aren't road-legal.

The rubber itself: it's not all the same 🧪

The rubber **compound** is the tire's secret: natural and synthetic rubber, silica, carbon and oils. The **soft** rubber grips more but wears out faster and gets hotter, while the **hard** rubber lasts longer but grips less. The maker balances **grip, life and fuel economy**, and Gulf heat wears soft rubber out faster.

Quality codes: UTQG 🔬

On the sidewall of some tires (especially in the US market) there's a rating system with three things:

Treadwear (a number like 400): A relative comparison against a reference tire rated 100 in a specific test. 400 means it's expected to last 4 times as long as the reference **in the test**. It is not a promise of kilometres.

Traction (AA/A/B/C): Straight-line braking on wet ground. AA is the highest.

Temperature (A/B/C): Resistance to heat build-up. A is the highest.

The system isn't uniform across markets: Europe uses a different label (wet grip, rolling resistance, noise), so don't compare one system with the other.

Tread and water: why the grooves? 🌧️

The grooves aren't decoration. Their first job is to **drain water** from under the tire so the rubber stays in contact with the tarmac, because water doesn't compress. Groove design varies with the goal:

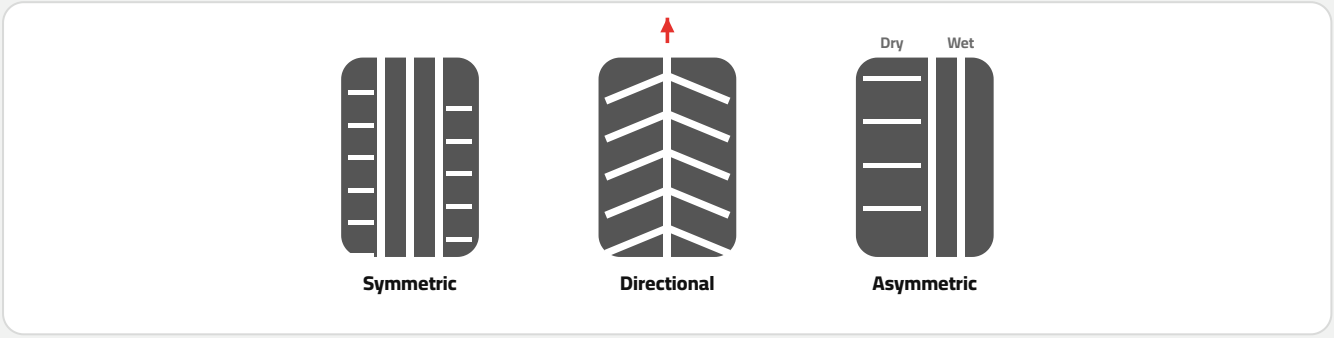
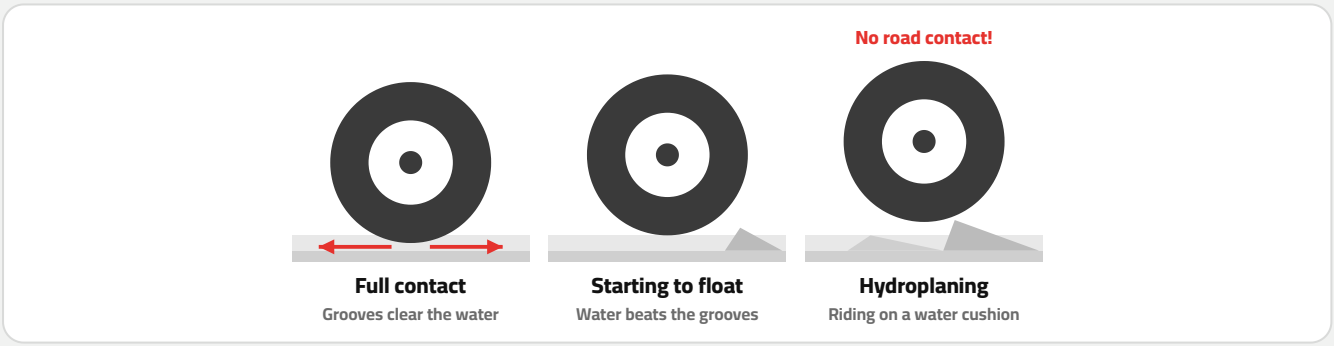


Illustration of three common designs, seen from above

Hydroplaning: when the tire rides up on the water

If the amount of water, the speed or the tread wear goes up, water builds up in front of the tire and gradually lifts it. At that moment steering and braking control drop sharply and can vanish completely, because **grip falls dramatically or disappears**.



Simplified drawing: the tire loses contact with the tarmac

Three things raise the risk: lots of water, high speed and a worn tread. Change the tire well before it reaches the minimum, especially if you drive where it rains.

A wider tire? It has to clear more water, so groove design and tread condition matter more than the width itself.

✕ MYTH
"The wider the tire, the faster the car."

✓ FACT
Width adds grip on dry roads, but it also adds drag, weight, fuel use and wet-weather risk. "Faster" is decided by the track, the load and the suspension, not by width alone.

Decoding the codes: 98Y and its relatives 🔓

In Chapter 1 we read the size 245/40R19. Right under the size you'll usually find two important codes. A full example:

245/40R19 98Y XL

98 = load index
A number that matches the maximum weight **one tire** can carry. A 98 means **750 kg** per tire.

Y = speed rating
The top speed the tire was designed to handle in the manufacturer's tests. Y is rated up to **300 km/h**.

XL
Extra Load: a reinforced tire that handles higher pressure and load than a standard tire of the same size.

Careful
The speed rating is the tire's capability, not permission to go that fast. Road limits always govern you.

Speed rating table (km/h) 🏁

Q	R	S	T	H	V	W	Y
160	170	180	190	210	240	270	300

Load index: how much can it carry? ⚖️

Index	85	90	95	**98**	100	105	110
kg per tire	515	600	690	**750**	800	925	1060

750 KG

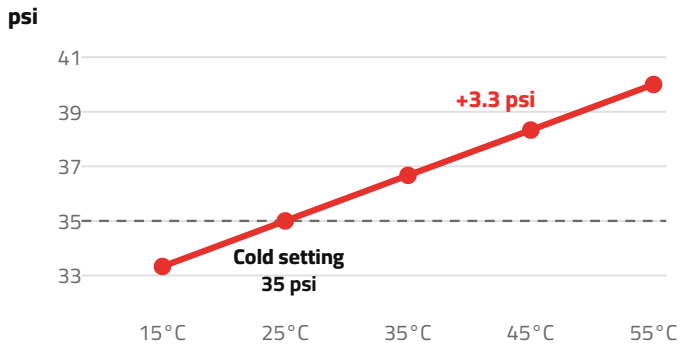
That's the maximum rated load of **one tire** with a load index of 98. Multiplying it by 4 doesn't give you the car's permitted payload; the reference is the car's label and owner's manual (gross weight and load per axle). Load index matters on loaded cars and SUVs, and any replacement size must be **equal to or higher than** the original specification, never lower.

🎯 Challenge 1 **In the code 98Y, what does the number 98 mean?**

A) Top speed | B) Load index (750 kg per tire) | C) Sidewall height

Pressure and heat: the most important number on your car 🌡️

The air inside the tire expands with heat. It's simple physics, but in the Gulf it carries real weight. The chart shows what happens to a tire set to 35 psi at 25 °C if the air temperature alone rises (even without driving):



Approximate calculation (gas law); in reality the tire heats up even more from driving and the asphalt

≈ 1 PSI

Roughly the rise for every 6 °C increase in air temperature. And the road surface here in summer can exceed 60 °C, so the tire heats up from below and from friction.

Check it cold

Before driving, or after about 3 hours parked in the shade. The correct figure is on the driver's door sticker (not the number on the tire sidewall, which is the maximum).

Don't let air out when hot

A rise of 2–4 psi after driving is normal. Letting air out while it's hot leaves you under-inflated once it cools.

Under-inflation is the bigger danger

It increases flexing and heat inside the sidewall, and is a known cause of tire blowouts in hot weather, especially with excess load.

Units

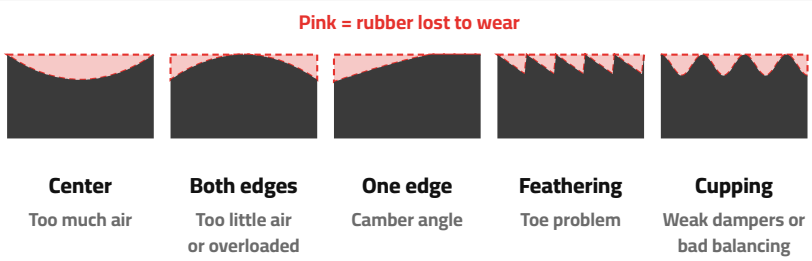
psi or bar or kPa. Roughly 1 bar = 14.5 psi.

🎯 Challenge 2 **When is the best time to measure tire pressure?**

A) After arriving from a long trip | B) While the tire is cold | C) At noon under the sun

Your tire talks: read the wear 🔎

Tread wear is like a medical report: it tells you where the problem is before it grows. Here are five common patterns (exaggerated so they're clear):



Cross-section of the tread in the first three, and a lengthwise section (around the tire's circumference) in the last two

Center wears faster

Too much air makes the middle bulge and take the friction.

Both edges wear

Too little air or overload, so the shoulders carry more than their share.

One edge (inner or outer)

Usually too much Camber angle, or damaged suspension.

A "feather" in one direction

You feel it when you run your hand across: smooth one way, sharp the other. Usually caused by Toe that isn't set right.

Cupping (patches)

Usually a weak damper or an unbalanced wheel.

> 💡 **Did you know?** Abnormal wear on one tire while the others are fine? The cause is usually not the tire itself but the suspension or the angles at that wheel.

Tire pressure monitoring (TPMS) 📡

Direct

A small sensor inside each wheel sends the actual pressure (and sometimes the temperature) to the car. Its battery has a limited life, and the technician has to handle it carefully when changing tires.

Indirect

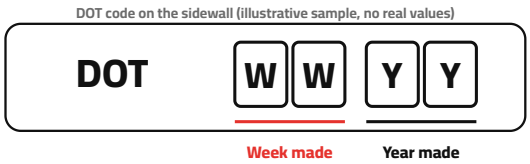
Uses the ABS wheel-speed sensors: an under-inflated tire turns slightly faster, and the car notices. Cheaper but less accurate.

✅ **DE.AUTO tip:** The warning light doesn't replace checking with a gauge: the system usually alerts only after a large drop (the percentage varies by car, system and country).

Tire age, rotation and balancing ⏳

Tires age even if you never drive them 🧓

Rubber oxidises and hardens with sun, heat and time, and Gulf heat speeds that up. A tire with “new tread but old manufacture” can crack at the sidewall.



The last four digits of the DOT code: two digits for the week and two for the year

The last four digits of the DOT code: two digits for the week (usually 01–52, and sometimes 53 in certain years) and two for the year.

Recommendations vary: many recommend a careful inspection at around 5 years, and not continuing with a tire more than about 10 years from manufacture even if it looks fine. Check your car’s manual.

The spare tire and stored tires age too: check the manufacture code when you buy.

Tire rotation 🔄

The front tires wear differently from the rear because they steer and brake. Rotation **spreads the wear** so the set lasts longer. The interval is usually between 8,000 and 10,000 km or per the manual, and the pattern differs if the tire is **directional** or the sizes differ front and rear.

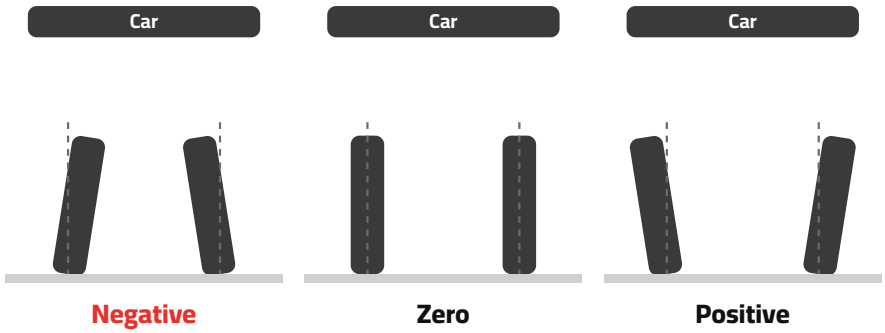
Wheel balancing ⚖️

A tire and wheel never have perfectly even weight distribution. The balancing machine spins the wheel to find the difference, and the technician adds **small weights** on the rim. Without it you feel **a vibration in the steering wheel or seat** at certain speeds.

✕ MYTH	✓ FACT
“Balancing and alignment are the same thing.”	Balancing deals with the **weight** of the wheel as it spins (it cures vibration). Alignment deals with the **direction** of the wheel relative to the car and the road (it cures pulling and wear). Two different jobs and two different machines.
✕ MYTH	✓ FACT
“Nitrogen makes the tire grip better and cuts heat significantly.”	Normal air is already 78% nitrogen. The difference in leakage is small, and the key is to check the pressure regularly whatever the gas.

Wheel angles: Camber 📐

Wheel alignment means setting each wheel's direction relative to the car and the road. There are three main angles: **Camber**, **Toe** and **Caster**. We start with the first:



Front view. The angle is exaggerated to show it. Negative: the top of the wheel leans toward the car

What is it?

The wheel's lean inward or outward when you look at it from the front of the car. Measured in degrees.

What does it change?

It changes the shape of the contact patch with the ground as the wheel leans with the body in a corner.

What changes in handling and wear?

Mild negative camber: Offsets body lean in a corner so more of the tire's surface stays on the ground in hard cornering. That's why many sports cars come with negative camber from the factory.

Too much negative: Reduces the contact patch in a straight line, so the **inner edge** of the tire wears faster, and braking distance may grow slightly.

Too much positive: Wears the **outer edge**, and stability and cornering control suffer.

Different angles left and right: The car pulls toward the side with the larger angle.

⚠️ **Important:** The correct camber figures are **set by the manufacturer for each model**, and they're in the service manual. Lowering the car and changing the suspension changes the camber automatically, which is why an alignment after any suspension modification isn't a luxury.

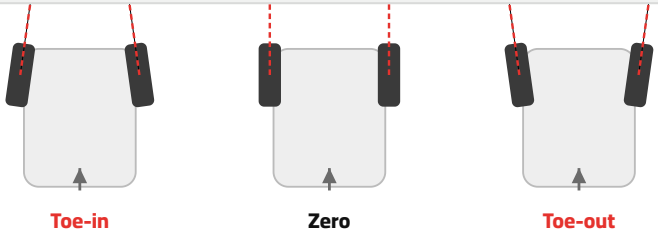
🎯 Challenge 3 **You find a tire wearing only on the inner edge and fine elsewhere. What's the most likely cause?**

A) Too much air | B) Too much negative camber | C) Missing balancing

Toe and Caster

Toe: where the nose points 👟

Look at the two wheels from above: are their fronts closer together or further apart? Like your feet when you walk with your toes pointing in or out. The angle is very small (millimetres or fractions of a degree).



Top view. The red lines show toe-in and toe-out (exaggerated)

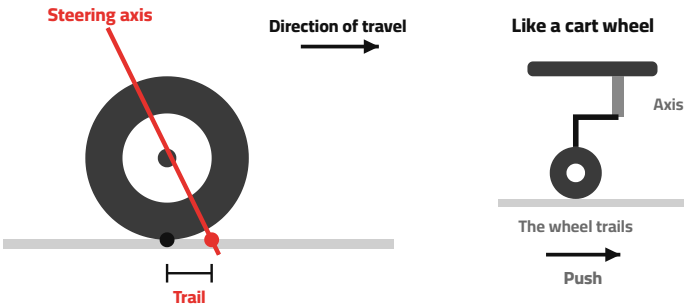
Slight toe-in: Adds stability in a straight line.

Slight toe-out: Improves turn-in response, but reduces stability.

Any excess toe: Makes the tire "drag" sideways instead of rolling straight, so it wears quickly in a feather pattern and raises fuel use.

Caster: like a shopping-cart wheel 🛒

Caster is the backward or forward tilt of the steering axis when you look at the car from the side. **Positive** caster (the most common) puts the axis point ahead of the contact point, so the wheel "trails" behind the axis like a shopping-cart wheel.



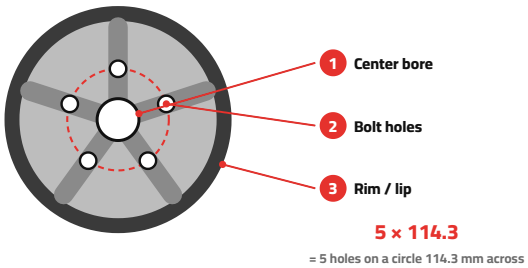
Side view. The red line: the steering axis. The trail gives stability

Caster gives you **steering that returns to centre** after a corner and stability in a straight line (more caster = heavier, steadier steering). It usually doesn't change tire wear directly, but a difference between left and right makes the car pull to one side.

🛠️ Setting all three angles is done on a special machine by a specialist technician, and it's redone after any change to the suspension or wheels.

Inside the wheel: reading its dimensions

Changing the wheel isn't only about diameter and looks. It has to fit your **hub** and tuck under your **wheel arch** without touching anything. These five numbers decide everything:



Front view of a typical wheel

Diameter and width

For example 19 × 8.5J: a 19-inch diameter and an 8.5-inch width (between the rim flanges).

PCD

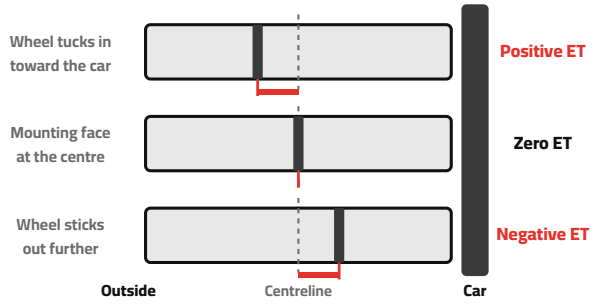
The number of bolt holes and the diameter of the circle they sit on, e.g. 5×114.3. It must match your car exactly.

Center bore

The middle hole. A better wheel sits on the hub precisely (hub-centric); otherwise it relies on the bolts alone (lug-centric).

Offset (ET)

The distance in millimetres between the mounting face and the wheel's centreline. It decides where the wheel sits inside the arch.



Top view of a wheel cross-section: the mounting face (black) moves with the offset

Backspacing: The distance from the mounting face to the inner edge of the wheel. Roughly = half the width in millimetres + the ET. Example: an 8.5-inch (216 mm) wheel with ET 35 → 108 + 35 ≈ 143 mm.

Challenge 4 **A wheel whose mounting face sits nearer the outside (higher ET). Where does the wheel end up?**

A) Sticks out further | B) Sits further in, toward the car | C) No change

Wheel materials and unsprung weight ⚙️

Cast
Aluminium poured into a mould. The most common and cheapest, and the heaviest.

Flow-formed
A cast centre, with the barrel formed by spinning and pressure so it ends up lighter and stronger.

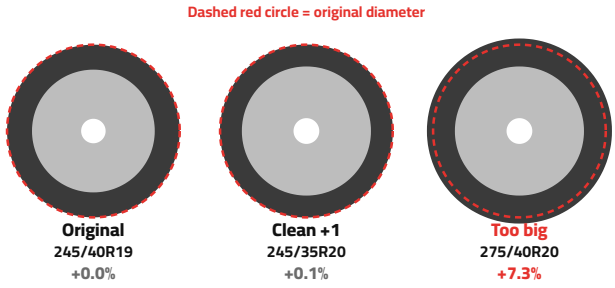
Forged
A single piece of aluminium shaped under huge pressure: lighter, stronger and more expensive, and the closest to track use.

Unsprung weight is everything the springs don't carry: wheels, tires, brakes and part of the suspension. The less there is, the easier it is for the suspension to follow bumps. (Steel wheels exist too: heavy, cheap and strong.)

> 🛡️ **Quality warning:** A no-name wheel at a tempting price can be **weaker** structurally. Look for recognised test marks (like JWL and VIA), and make sure the wheel's load rating suits your car.

Upsizing the wheel: the maths of plus-sizing 🧮

The rule: **the wheel gets bigger and the sidewall gets smaller, and the overall diameter stays roughly the same.** Overall diameter = (sidewall height × 2) + (rim diameter × 25.4).



Percentages = difference in overall diameter from the original

Original 245/40R19: sidewall height 245 × 0.40 = 98 mm, so the diameter is 2×98 + 19×25.4 = **678.6** mm.

245/35R20: 171.5 + 508 = **679.5** mm (+0.13%). Acceptable.

275/40R20: 220 + 508 = **728** mm (+7.3%). Too big!

+7.3%

If the speedometer shows 100 km/h, your real speed is about **107 km/h**, and the odometer under-counts by the same percentage. It also confuses ABS and stability systems and changes the effective gear ratio.

Fitment problems: rubbing, spacers and stretch ⚠️

Rubbing

When the tire touches the fender, the liner or the suspension arms over bumps or on full lock. Causes: a tire wider than designed, an unsuitable offset, a larger diameter, or a lowered car. The danger isn't only the noise: repeated rubbing cuts the tire's sidewall and it blows out suddenly.

Spacers / adapters

A spacer is a part between the hub and the wheel that pushes the wheel outward, and the adapter also changes the bolt pattern (PCD). People use them to line the wheel up with the arch or to clear a big caliper. **But they carry real risks:**

Hub centring
It must centre on the hub precisely (hub-centric), otherwise you get vibration and stress on the studs.

Stud length
You must keep enough thread engagement inside the nut after adding the thickness, and too little is dangerous.

Quality and load
Poor material means cracking or loosening, and pushing the wheel outward adds stress to the wheel bearings and suspension.

✅ **The safer way:** Choose a wheel with the right offset from the start instead of patching with a spacer, and if you must, have a qualified technician decide, inspect and re-torque per the manufacturer's recommendation.

Stretched tire 😬

✕ MYTH
"A narrow tire on a wide wheel looks great and does no harm."

✓ FACT
Manufacturers specify a range of wheel widths for each tire size. Outside it, the sidewall is exposed to impacts, the tire bead can come off the wheel in a bump or corner, and stability can suffer. This is one of the most dangerous street "styles".

The legal side and inspection 📋

A minimum tread depth is usually required (about 1.6 mm in most systems), and the tire must not be damaged, bulging or cracked.

A tire or wheel sticking out past the arch, or a size that throws off speedometer accuracy and safety, may count as a violation or fail the periodic inspection.

Rules differ from country to country: check with the official authority before any change.

Common mistakes that cost you ❌

❌ **Buying a tire on price alone:** You save SAR 500 and lose braking distance, stability and safety. The tire is the last place to save.

❌ **Mixing different tires on the same axle:** Changes handling and braking. And on all-wheel drive, a difference in tire diameter strains the drivetrain. Change all four, or as the manufacturer recommends.

❌ **Ignoring vibration or the car pulling:** These are early warnings. Wait, and you wear out a tire and damage the suspension.

❌ **Not checking pressure in summer:** Heat changes the pressure. A monthly check with an accurate gauge is cheaper than any repair.

❌ **Upsizing the wheel without doing the diameter maths:** You pay for the wheel and get a wrong speedometer and a rubbing tire. And a bigger wheel "isn't always better": it cuts comfort and exposes the wheel to potholes.

Garage talk vs. the proper name 🗣️

Gulf garage talk	Proper name	What it means
Jant	Wheel (or rim)	Technically the rim is just the edge, but the common name covers the whole wheel.
Kafar	Tire	From the word "cover"; some regions say "kootch".
Tarsees / wazn	Wheel balancing	Corrects weight distribution to reduce vibration. The name varies by region.
Dabt zawaya	Wheel alignment	Camber, Toe and Caster.
Banshar	Puncture repair	The tire shop.
Tabdeel al-kafarat	Tire rotation	Evens out the wear.
Masameer al-jant	Wheel bolts / nuts (Lug Nuts / Studs)	Fasten the wheel to the hub.
Fasil / mubaaid	Wheel Spacer	A part that pushes the wheel outward.
Offset	ET (Offset)	Where the mounting face sits.
Lambet al-hawa	TPMS	Tire pressure warning.

Wrap-up 📌

Now you know your whole car stands on **four patches** the size of a palm: a tire of the right type at the right pressure, aligned angles, and a wheel with the right size and offset. This is the foundation before any other modification. 🛞

🎯 Challenge answers:

1) B: Load index, 98 = 750 kg per tire.

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2) B: While the tire is cold.

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3) B: Too much negative camber.

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4) B: Further in toward the car (higher ET = the wheel sits deeper).

➡️ In **Chapter 11: Aerodynamics and Weight Reduction**, we look at how the air around you acts and how weight changes everything: wings, diffusers, streamlining, and why every kilo matters. Ready? 🚀